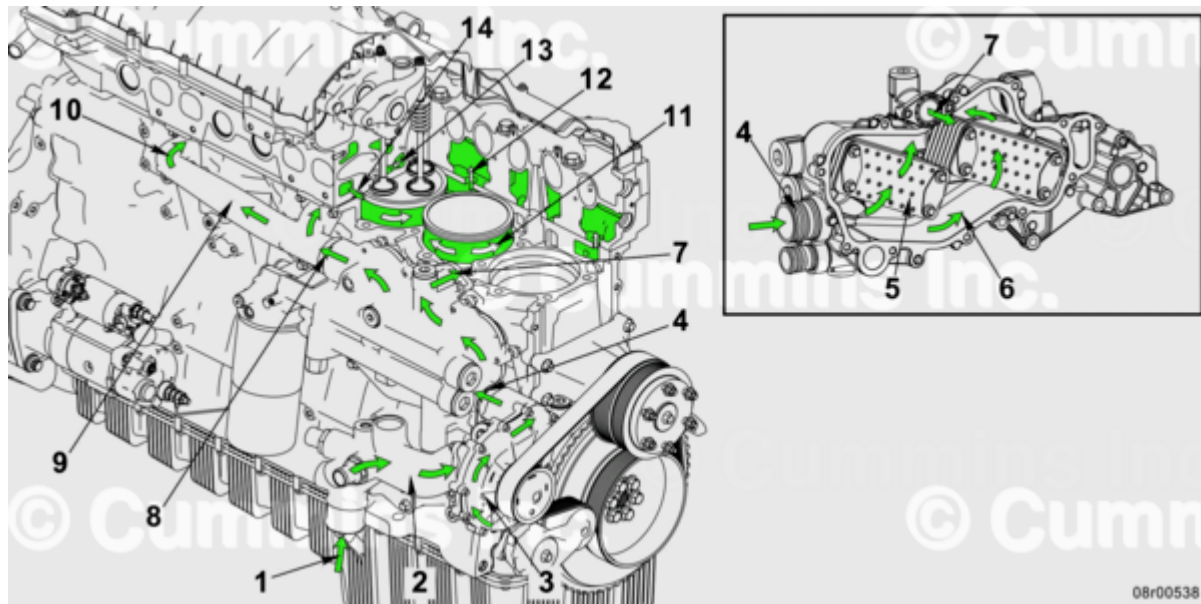


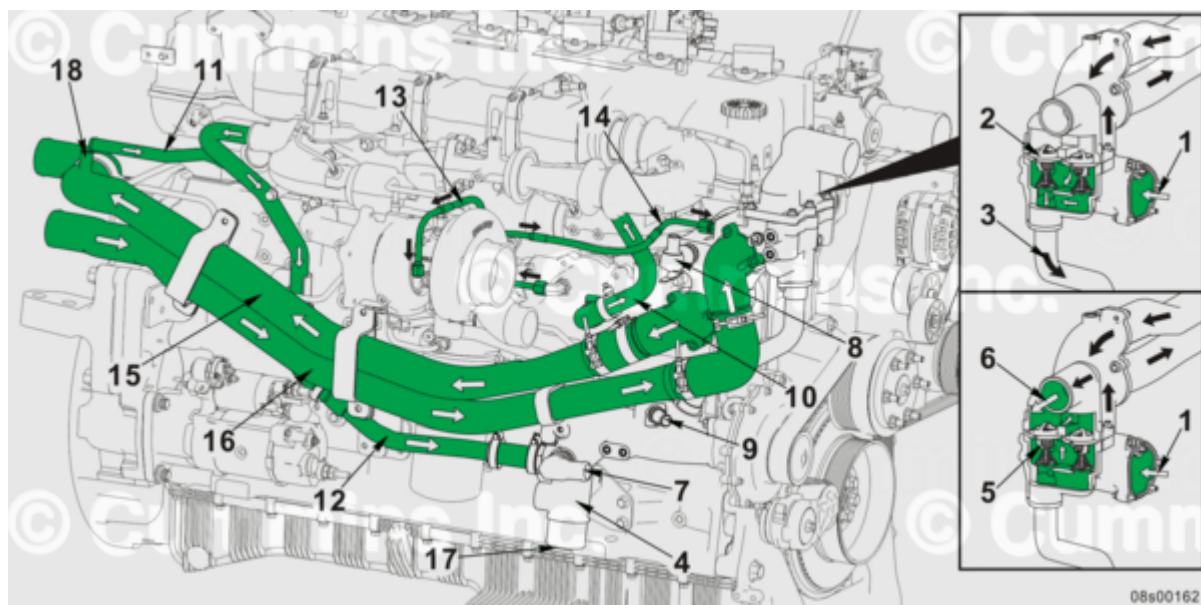
Trainings ▾

Flow Diagram



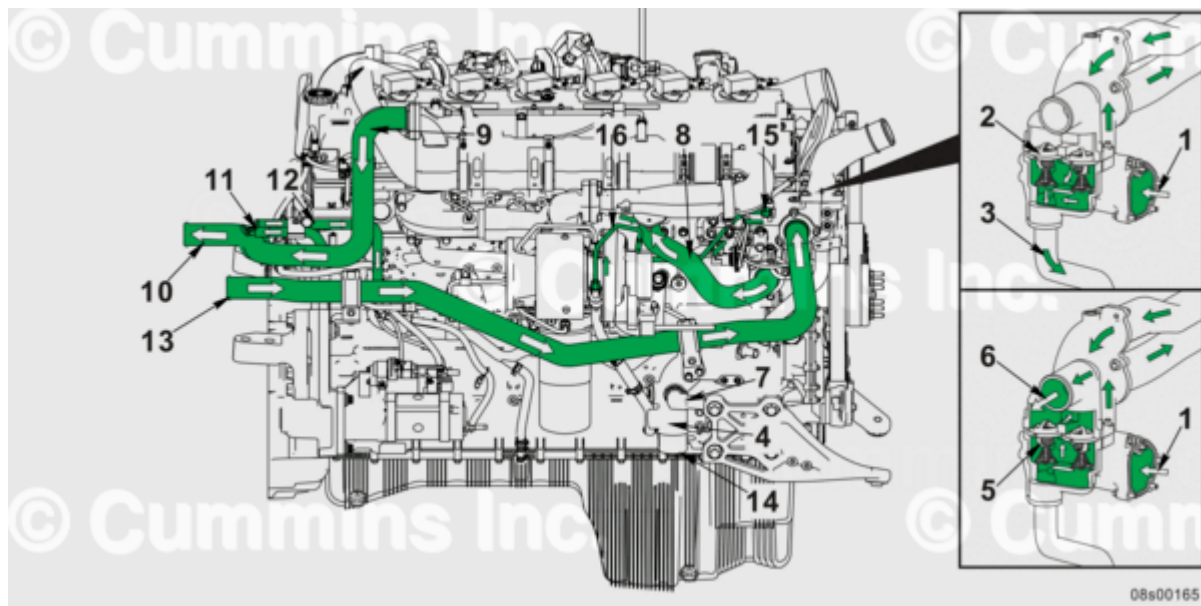
Flow Through Engine

1. Coolant from radiator
2. Coolant flow to water pump
3. Water pump
4. Coolant flow from water pump to oil cooler module
5. Oil cooler element
6. Coolant flow around oil cooler element
7. Coolant flow from oil cooler module to cylinder block
8. Coolant flow from oil cooler module to coolant manifold
9. Coolant manifold
10. Coolant flow from coolant manifold to cylinder block
11. Coolant flow around cylinders
12. Coolant flow from cylinder block to lower cylinder head
13. Coolant flow to upper cylinder head
14. Coolant flow from cylinder head to thermostat housing.



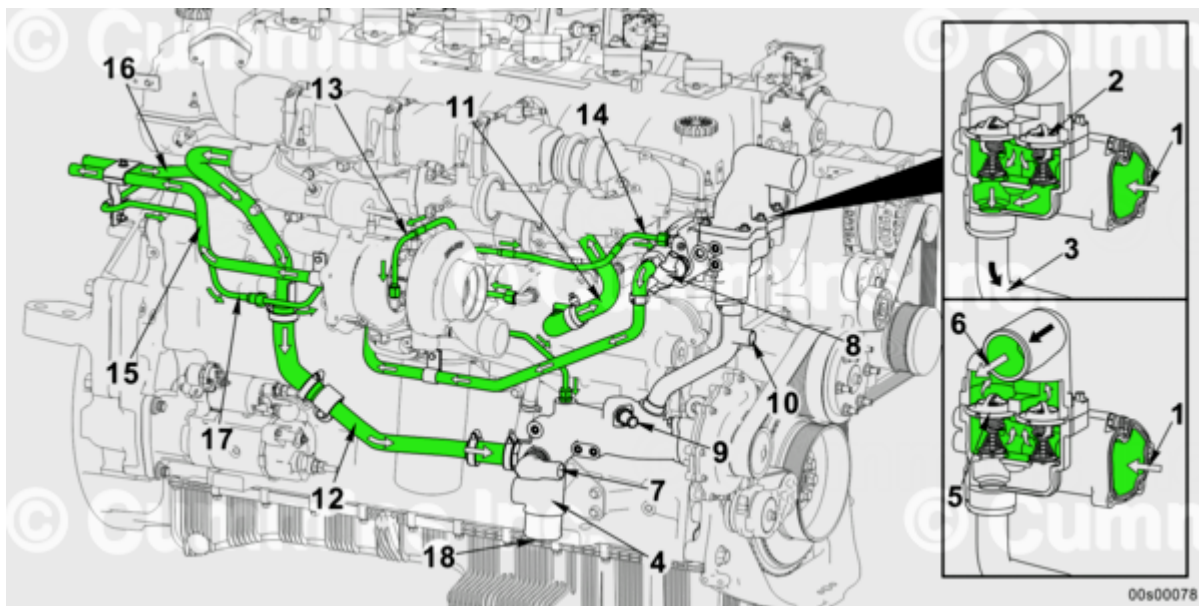
Flow Through Coolant Thermostat - With Retarder, Parallel

1. Coolant flow from cylinder head
2. Thermostat closed
3. Coolant flow through bypass tube to coolant inlet connection
4. Coolant inlet connection
5. Thermostat open
6. Coolant flow to radiator
7. Coolant flow return from water tank (optional)
8. Coolant flow to cabin heater
9. Coolant flow return from cabin heater
10. Coolant from lubricating oil cooler to Exhaust Gas Recirculation (EGR) cooler
11. Coolant from evaporator (Liquefied Natural Gas (LNG)) or regulator (Compressed Natural Gas (CNG))
12. Coolant flow return from EGR cooler
13. Coolant from lubricating oil cooler to turbocharger bearing housing
14. Coolant flow return from turbocharger bearing housing
15. Coolant flow to retarder plumbing
16. Coolant flow return from retarder plumbing
17. Coolant flow from radiator
18. Coolant flow to evaporator (LNG) or regulator (CNG).



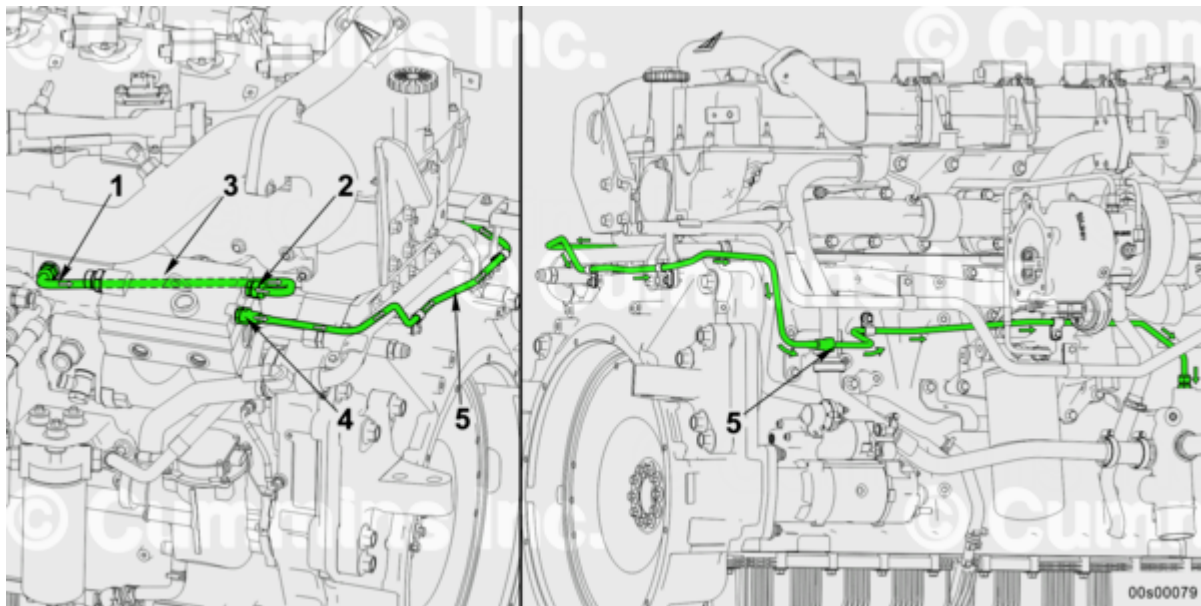
Flow Through Coolant Thermostat - With Retarder, Series

1. Coolant flow from cylinder head
2. Thermostat closed
3. Coolant flow through bypass tube to coolant inlet connection
4. Coolant inlet connection
5. Thermostat open
6. Coolant flow to radiator
7. Coolant flow return from water tank (optional)
8. Coolant flow to EGR cooler
9. Coolant flow through EGR cooler to retarder plumbing
10. Coolant flow through retarder plumbing to retarder
11. Coolant flow through retarder plumbing to evaporator (LNG) or regulator (CNG)
12. Coolant flow from evaporator (LNG) or regulator (CNG)
13. Coolant flow from retarder
14. Coolant flow from radiator
15. Coolant from thermostat housing to turbocharger bearing housing
16. Coolant flow return from turbocharger bearing housing.



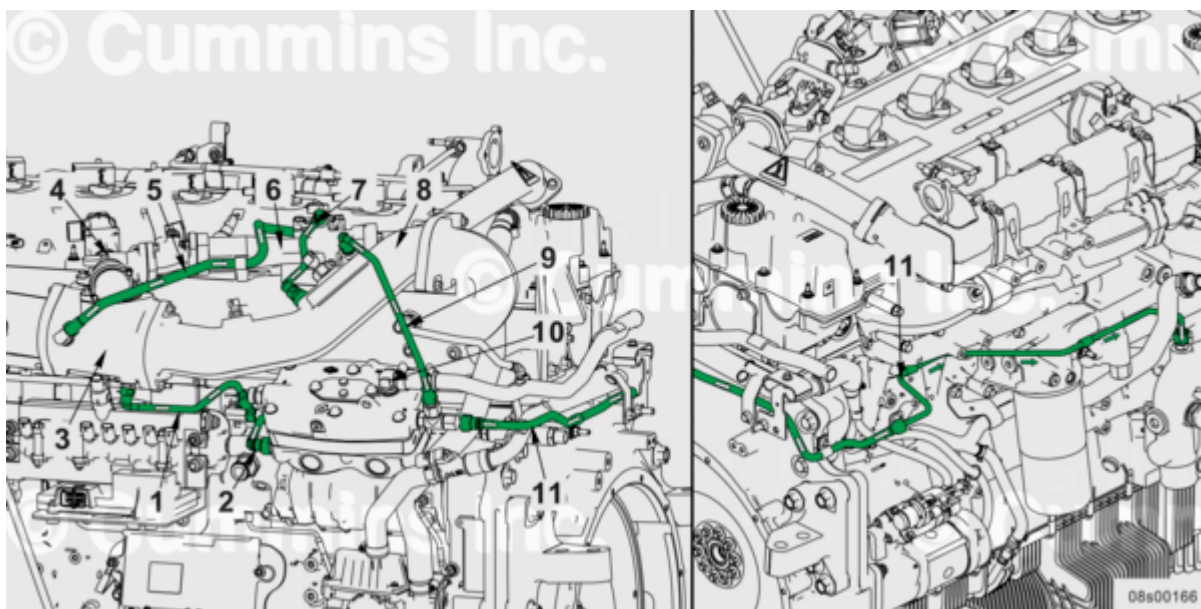
Flow Through Coolant Thermostat - Without Retarder

1. Coolant flow from cylinder head
2. Thermostat closed
3. Coolant flow through bypass tube to coolant inlet connection
4. Coolant inlet connection
5. Thermostat open
6. Coolant flow to radiator
7. Coolant flow return from water tank (optional)
8. Coolant flow to cabin heater
9. Coolant flow return from cabin heater
10. Coolant flow return from water tank (optional)
11. Coolant from lubricating oil cooler to EGR cooler
12. Coolant flow return from EGR cooler
13. Coolant from lubricating oil cooler to turbocharger bearing housing
14. Coolant flow return from turbocharger bearing housing
15. Coolant flow to evaporator (LNG) or regulator (CNG)
16. Coolant from evaporator (LNG) or regulator (CNG)
17. Coolant flow from air compressor return line to water inlet connection
18. Coolant flow from radiator.



Flow Through Air Compressor, Without EGR Water Jacket

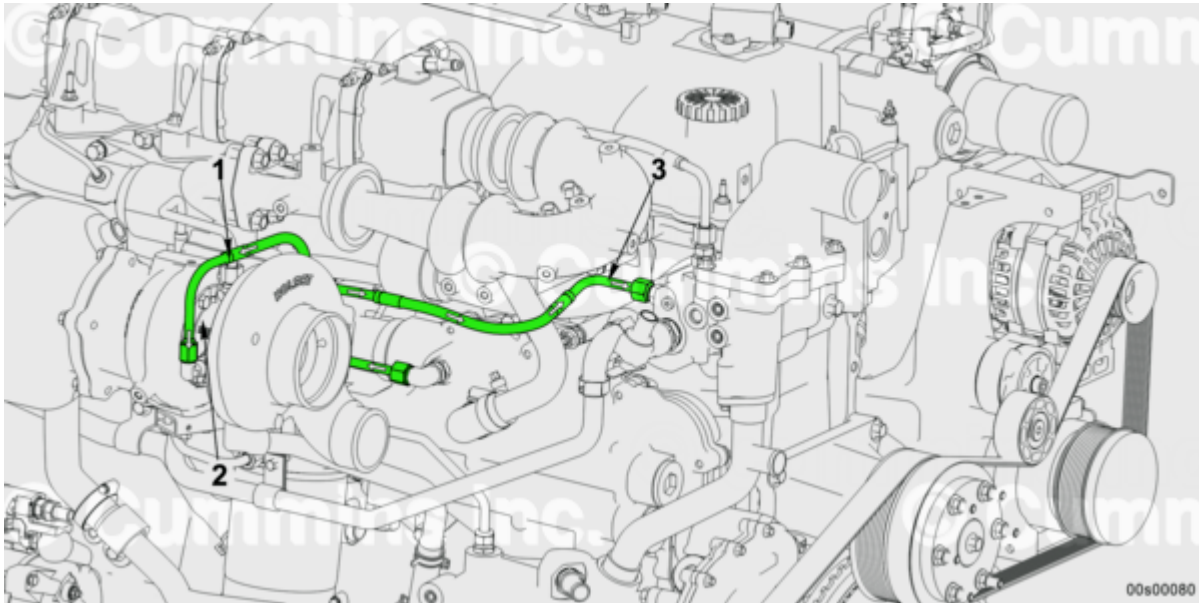
1. Coolant flow from cylinder block to air compressor
2. Coolant inlet port
3. Air compressor
4. Coolant outlet port
5. Coolant flow from air compressor return line to water inlet connection.



Flow Through EGR/Air Intake System, With Water Jacket

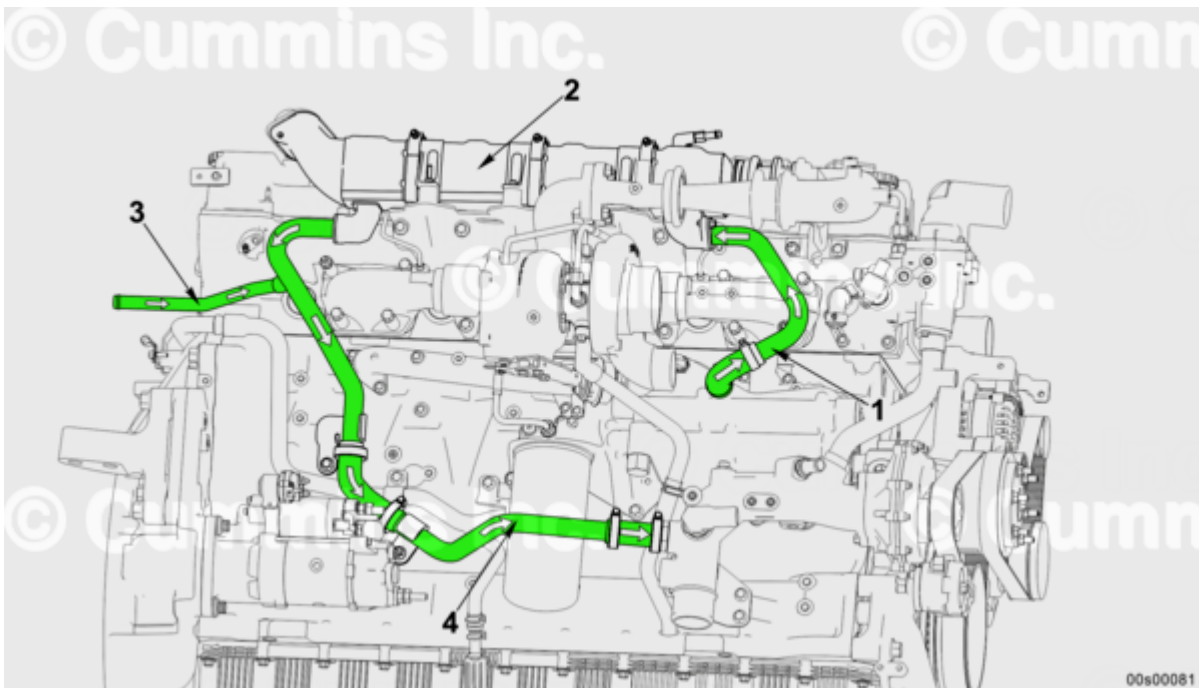
1. Coolant flow from cylinder block to EGR mixer
2. Coolant flow from cylinder block to air compressor
3. EGR mixer
4. EGR valve
5. Coolant flow from EGR mixer to venturi tube
6. Venturi tube
7. Coolant flow from venturi tube to air intake connection

8. Air intake connection
9. Coolant flow from air intake connection to water inlet connection
10. Air compressor
11. Coolant flow from air compressor to water inlet connection.



Flow Through Turbocharger

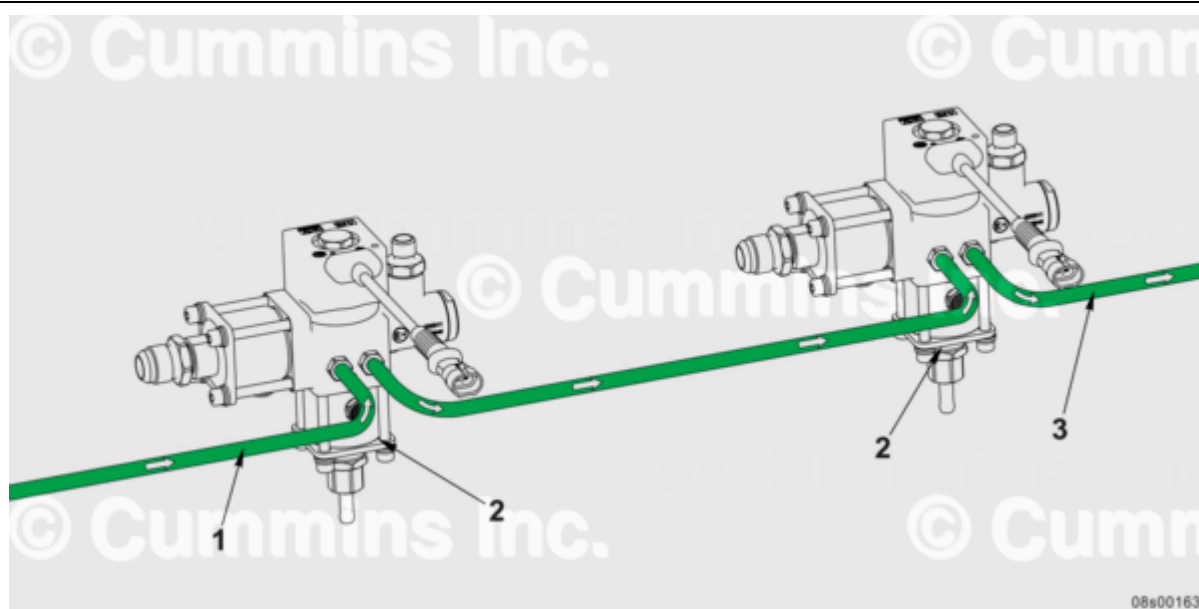
1. Coolant from lubricating oil cooler to turbocharger bearing housing
2. Turbocharger bearing housing
3. Coolant from turbocharger bearing housing to coolant thermostat housing.



Flow Through EGR Cooler

1. Coolant from lubricating oil cooler to EGR cooler
2. EGR cooler

3. Coolant from evaporator (LNG) or regulator (CNG)
4. Coolant from EGR cooler to water inlet connection.



Flow Through Regulators (CNG)

1. Coolant to fuel pressure regulator (CNG)
2. Fuel pressure regulator (CNG)
3. Coolant from fuel pressure regulator (CNG).